Nicolò Bonacorsi

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Education

Columbia University — New York, NY
MS in Applied Mathematics, GPA: 3.9/4.00 — Aug 2025 – Dec 2026

Sapienza University of Rome — Rome, Italy
BS in Mathematical Sciences for AI, GPA: 3.99/4.00 | 110/110 *summa cum laude* — Sep 2022 – Sep 2025
Achievement: Earned two B.S. degrees with highest honors, pursuing both concurrently at separate universities.

University of Bologna — Bologna, Italy
BS in Statistical Sciences, GPA: 3.95/4.00 | 110/110 *summa cum laude* — Sep 2021 – Jul 2024
Award: University of Bologna Prize for Outstanding Student

Research Experience

Columbia University | Supervisor: Professor **Ioannis Karatzas** — New York, NY
Graduate Researcher — Dec 2025 – Present
Drift and Volatility Control with Discretionary Stopping – Working paper.
- Derived optimal stopping thresholds and proved constant drift and dispersion controls on continuation tails for convex terminal costs; characterized nonconvex free boundaries and optimal state-dependent policies for multiple quadratic targets.
- Implemented a Python free-boundary solver validated by finite differences and Monte Carlo across 132 regimes; joint control cut cost by up to **15.3%**, with **0.37%** worst-case regret under ±10% misspecification.

Northeastern University | Supervisor: Professor **Francesco Restuccia** — Boston, MA
Research Collaborator — Dec 2025 – Present
Credal Rate–Distortion Theory – *IEEE ITW 2026*; *extended version under review at IEEE Trans. Inf. Theory.*
N. Bonacorsi*, M. Caprio, F. Meneghello, F. Restuccia.*
- Formulated a robust rate–distortion theory for ML under model uncertainty; proved zero-distortion credal compression is exactly graph coloring of an incompatibility graph.

The Credal Information Bottleneck – submitted to ICLR 2027.
D. Loi, G. Onorato, ***N. Bonacorsi****, F. Meneghello, F. Restuccia.*
- Generalized information-bottleneck optimization to sets of plausible laws; across three benchmarks, cut representation rate **35%** at matched worst-case accuracy and gained **+6 pp** accuracy at matched rate versus standard and robust IB baselines.

Work Experience

MENTIS Laboratory — Boston, MA
AI Research Intern — Jul 2026 – Aug 2026
- Built a Python LP-feasibility/graph-search optimizer; solved 54,000 problems **20**× faster and within 1% of the exact optimum.
- Stress-tested uncertain-ML codebooks; compressed 60 uncertainty sets to **14 states (4.29×)** with zero violations and near-zero violations under 15% parameter perturbations.

E.PTY Group — Bologna, Italy
Quantitative Researcher — Jun 2023 – Jun 2025
- Built hierarchical/posterior-predictive Monte Carlo allocation; lowered rolling OOS RMSE **14.6%** versus historical baselines.
- Deployed rolling stochastic allocation with downside constraints; raised **realized operating profit by 11.8%**, while historical backtests lowered 95% downside CVaR by **17.3%**.

Quantitative Projects

Certified Alpha Capacity (AlphaValue) [arXiv] [GitHub] [Zenodo] — Aug 2026 – Present
- Built and released *AlphaValue*, a public alpha-certification research package; derived exact multiple-testing feasibility thresholds and a crowding equilibrium linking arbitrage entry to endogenous signal decay.
- Validated *AlphaValue* on 543 BTC/ETH funding events: lifetime score beat current edge (**0.476 vs. 0.383**) and retained **0.282** incremental rank signal (95% CI [**0.128,0.412**]).

Maturity-Safe Distributional Volatility Forecasting [GitHub] — May 2026 – Jul 2026
- Assembled a leakage-safe 22-day variance forecaster using six experts, bounded-loss Hedge, maturity-delayed updates, and 22 non-overlapping residual streams.
- Evaluated 3,381 OOS forecasts, cutting log-RMSE **14.9% vs. VIX** (HAC $p < 0.001$) with 89.0% coverage at nominal 90%; a volatility-targeting controller cut max drawdown to **11.6% vs. 15.3%** for fixed exposure.

Bootstrap-Calibrated Distributionally Robust Allocation [GitHub] — May 2026 – Jul 2026
- Designed a DRO allocator for 25 Fama–French portfolios using circular-block-bootstrap covariance uncertainty and capped long-only QPs with a custom active-set solver.
- Achieved **16.45%** OOS volatility and **0.820** net Sharpe after 10 bps, matching Ledoit–Wolf GMV while cutting volatility **18.4%** versus equal weight.

Bayesian Modeling of Collatz Stopping Times [arXiv] — 2026
- Modeled stopping-time distributions over 10M integers via hierarchical Bayes and Monte Carlo; regime conditioning improved Wasserstein fit by **3.2**×.

Skills

Programming & Data: Python, SQL, R, MATLAB; NumPy, pandas, SciPy, scikit-learn, PyTorch, statsmodels.
Methods: Probability, Bayesian Inference, Time Series, Monte Carlo, Machine Learning, Stochastic Processes, Stochastic Control, Optimal Stopping, Optimization, Backtesting, Risk Modeling, Portfolio Optimization.
Languages: Italian (native), English (fluent).